

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2021 Worked Solutions

January 2021 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 January 2021

January 2021 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Polynomials

1.
$$f(x) = x^4 + ax^3 - 3x^2 + bx + 5$$

where a and b are constants.

When $f(x)$ is divided by $(x + 1)$, the remainder is 4

(a) Show that $a + b = -1$

(2)

When $f(x)$ is divided by $(x - 2)$, the remainder is -23

(b) Find the value of a and the value of b .

(4)

Worked Solution - Question 1

1. Use the first remainder

Dividing by $(x + 1)$ means use $x = -1$. Hence

$$f(-1) = 1 - a - 3 - b + 5 = 3 - a - b.$$

2. Show the required relation

The remainder is 4, so $3 - a - b = 4$. Therefore $a + b = -1$.

3. Use the second remainder

Dividing by $(x - 2)$ means use $x = 2$. Hence

$$f(2) = 16 + 8a - 12 + 2b + 5 = 9 + 8a + 2b.$$

4. Form the second equation

The remainder is -23 , so $9 + 8a + 2b = -23$, which gives $4a + b = -16$.

5. Solve simultaneously

Subtract $a + b = -1$ from $4a + b = -16$ to get $3a = -15$, so $a = -5$.

6. Find b

Using $a + b = -1$, $-5 + b = -1$, hence $b = 4$.

Final answer

(a) Proven: $a + b = -1$.

(b) $a = -5$, $b = 4$.

Question 2

Applications of Differentiation

2. A curve has equation

$$y = x^3 - x^2 - 16x + 2$$

- (a) Using calculus, find the x coordinates of the stationary points of the curve. (4)
- (b) Justify, by further calculus, the nature of all of the stationary points of the curve. (3)

Worked Solution - Question 2

Topic group

1. Differentiate the curve

For $y = x^3 - x^2 - 16x + 2$, $\frac{dy}{dx} = 3x^2 - 2x - 16$.

2. Set the gradient to zero

Stationary points occur when $3x^2 - 2x - 16 = 0$.

3. Factorise

$$3x^2 - 2x - 16 = (3x - 8)(x + 2), \text{ so } x = \frac{8}{3} \text{ or } x = -2.$$

4. Differentiate again

$$\frac{d^2y}{dx^2} = 6x - 2.$$

5. Classify x equals negative two

At $x = -2$, $6x - 2 = -14 < 0$, so the stationary point is a local maximum.

6. Classify x equals eight over three

At $x = \frac{8}{3}$, $6x - 2 = 14 > 0$, so the stationary point is a local minimum.

Final answer

(a) $x = -2$, $\frac{8}{3}$. (b) local maximum at $x = -2$ and local minimum at $x = \frac{8}{3}$.

Question 3

Laws of Logarithms

3. (i) Solve

$$7^{x+2} = 3$$

giving your answer in the form $x = \log_7 a$ where a is a rational number in its simplest form.

(3)

- (ii) Using the laws of logarithms, solve

$$1 + \log_2 y + \log_2 (y + 4) = \log_2 (5 - y)$$

(5)

Worked Solution - Question 3

1. Rewrite the exponential equation

From $7^{x+2} = 3$, divide by 7^2 to get $7^x = \frac{3}{49}$.

2. Write x as a logarithm

Therefore $x = \log_7 \left(\frac{3}{49} \right)$.

3. Convert one to a logarithm

Since $1 = \log_2 2$, the equation becomes
 $\log_2 2 + \log_2 y + \log_2(y + 4) = \log_2(5 - y)$.

4. Combine the logarithms

$\log_2\{2y(y + 4)\} = \log_2(5 - y)$.

5. Solve the quadratic

Thus $2y(y + 4) = 5 - y$, so $2y^2 + 9y - 5 = 0 = (2y - 1)(y + 5)$.

6. Check the domain

The original logarithms require $y > 0$ and $y < 5$, so $y = -5$ is invalid and $y = \frac{1}{2}$.

Final answer

(i) $x = \log_7 \left(\frac{3}{49} \right)$. (ii) $y = \frac{1}{2}$.

Question 4

Binomial Expansion

4. (a) Find the first three terms, in ascending powers of x , of the binomial expansion of

$$(2 + px)^6$$

where p is a constant. Give each term in simplest form.

(4)

Given that in the expansion of

$$\left(3 - \frac{1}{2}x\right)(2 + px)^6$$

the coefficient of x^2 is $-\frac{3}{4}$

- (b) find the possible values of p .

(4)

Worked Solution - Question 4

1. Write the first binomial terms

$$(2 + px)^6 = 2^6 + \binom{6}{1}2^5(px) + \binom{6}{2}2^4(px)^2 + \dots$$

2. Simplify the expansion

The first three terms are $64 + 192px + 240p^2x^2$.

3. Find the coefficient of x squared

In $(3 - \frac{1}{2}x)(2 + px)^6$, the x^2 coefficient is $3(240p^2) - \frac{1}{2}(192p)$.

4. Form the equation

$$\text{So } 720p^2 - 96p = -\frac{3}{4}.$$

5. Solve the quadratic

Multiplying by 4 gives $2880p^2 - 384p + 3 = 0$, or $960p^2 - 128p + 1 = 0$.

6. State the values of p

$$p = \frac{128 \pm 112}{1920}, \text{ so } p = \frac{1}{8} \text{ or } p = \frac{1}{120}.$$

Final answer

$$(a) 64 + 192px + 240p^2x^2. \quad (b) p = \frac{1}{8} \text{ or } p = \frac{1}{120}.$$

Question 5

Proof

5. (i) Use algebra to prove that for all $x \geq 0$

$$3x + 1 \geq 2\sqrt{3x} \quad (3)$$

- (ii) Show that the following statement is not true.

“The sum of three consecutive prime numbers is always a multiple of 5”
(1)

Worked Solution - Question 5

Topic group

1. Start from a square

For $x \geq 0$, $\sqrt{3x}$ is real and $(\sqrt{3x} - 1)^2 \geq 0$.

2. Expand the square

$$(\sqrt{3x} - 1)^2 = 3x - 2\sqrt{3x} + 1 \geq 0.$$

3. Rearrange the inequality

Therefore $3x + 1 \geq 2\sqrt{3x}$, as required.

4. Give a counterexample

The consecutive prime numbers **5**, **7**, and **11** have sum $5 + 7 + 11 = 23$.

5. Complete the disproof

23 is not a multiple of **5**, so the statement is not true.

Final answer

(i) Proven.

(ii) For example, $5 + 7 + 11 = 23$, which is not a multiple of **5**.

Question 6

Trigonometric Equations

6. (a) Show that the equation

$$\frac{3\sin\theta\cos\theta}{2\sin\theta-1} = 5\tan\theta \quad \sin\theta \neq \frac{1}{2}$$

can be written in the form

$$3\sin^3\theta + 10\sin^2\theta - 8\sin\theta = 0 \quad (4)$$

- (b) Hence solve, for $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\frac{3\sin 2x\cos 2x}{2\sin 2x-1} = 5\tan 2x$$

giving your answers to 3 decimal places where appropriate. (4)

Worked Solution - Question 6

1. Clear the fraction

Starting with $\frac{3 \sin \theta \cos \theta}{2 \sin \theta - 1} = 5 \tan \theta$, multiply by $(2 \sin \theta - 1) \cos \theta$.

2. Use tan theta

This gives $3 \sin \theta \cos^2 \theta = 5 \sin \theta (2 \sin \theta - 1)$.

3. Use cos squared

Substitute $\cos^2 \theta = 1 - \sin^2 \theta$: $3 \sin \theta (1 - \sin^2 \theta) = 10 \sin^2 \theta - 5 \sin \theta$.

4. Collect terms

Rearranging gives $3 \sin^3 \theta + 10 \sin^2 \theta - 8 \sin \theta = 0$.

5. Factor the sine equation

$3s^3 + 10s^2 - 8s = 0$ gives $s(3s - 2)(s + 4) = 0$, where $s = \sin 2x$.

6. Use the valid sine values

Since $-\frac{\pi}{2} < 2x < \frac{\pi}{2}$, the valid values are $\sin 2x = 0$ and $\sin 2x = \frac{2}{3}$.

7. Find x

$2x = 0$ gives $x = 0$. Also $2x = \sin^{-1}\left(\frac{2}{3}\right)$, so $x = 0.364863 \dots = 0.365$ to 3 decimal places.

Final answer

(*a*) Proven.

(*b*) $x = 0, 0.365$.

Question 7

Integration

7.

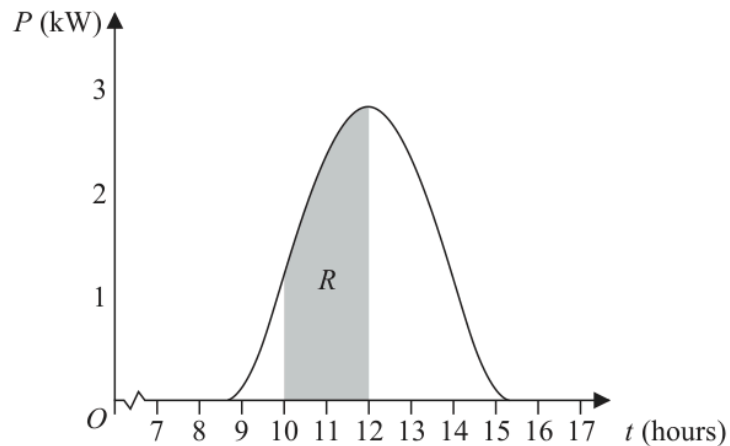


Figure 1

Solar panels are installed on the roof of a building.

The power, P , produced on a particular day, in kW, can be modelled by the equation

$$P = 0.95 + 2^{t-12} + 2^{12-t} - (t-12)^2 \quad 8.5 \leq t \leq 15.2$$

where t is the time in hours after midnight. The graph of P against t is shown in Figure 1.

A table of values of t and P is shown below, with the values of P given to 4 significant figures where appropriate.

Time, t (hours)	10	10.5	11	11.5	12
Power, P (kW)		1.882	2.45		2.95

- (a) Use the given equation to complete the table, giving the values of P to 4 significant figures where appropriate.

(2)

The amount of energy, in kWh, produced between 10:00 and 12:00 can be found by calculating the area of region R , shown shaded in Figure 1.

- (b) Use the trapezium rule, with all the values of P in the completed table, to find an estimate for the amount of energy produced between 10:00 and 12:00. Give your answer to 2 decimal places.

(4)

Worked Solution - Question 7

1. Use the model at t equals ten

$$P(10) = 0.95 + 2^{-2} + 2^2 - (10 - 12)^2 = 0.95 + 0.25 + 4 - 4 = 1.20.$$

2. Use the model at t equals eleven point five

$$P(11.5) = 0.95 + 2^{-0.5} + 2^{0.5} - (11.5 - 12)^2 = 2.821 \dots$$

3. Identify the trapezium width

The time values increase by 0.5, so $h = 0.5$.

4. Apply the trapezium rule

The estimate is $\frac{0.5}{2} \{1.20 + 2.95 + 2(1.882 + 2.45 + 2.821 \dots)\}$.

5. Calculate the energy

This gives 4.614..., so the estimated energy is **4.61 kWh** to 2 decimal places.

Final answer

(a) $P(10) = 1.20$, $P(11.5) = 2.821$. (b) 4.61 kWh.

Question 8

Sequences & Series

8. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = 2(a_n + 3)^2 - 7$$
$$a_1 = p - 3$$

where p is a constant.

- (a) Find an expression for a_2 in terms of p , giving your answer in simplest form.

(1)

Given that $\sum_{n=1}^3 a_n = p + 15$

- (b) find the possible values of a_2

(6)

Worked Solution - Question 8

1. Find a two

Since $a_1 = p - 3$, we have $a_1 + 3 = p$. Therefore

$$a_2 = 2(a_1 + 3)^2 - 7 = 2p^2 - 7.$$

2. Find a three

$$a_2 + 3 = 2p^2 - 4, \text{ so } a_3 = 2(2p^2 - 4)^2 - 7 = 8(p^2 - 2)^2 - 7.$$

3. Use the given sum

$$a_1 + a_2 + a_3 = p + 15.$$

4. Substitute the expressions

$$(p - 3) + (2p^2 - 7) + 8(p^2 - 2)^2 - 7 = p + 15.$$

5. Simplify

Cancelling p and expanding gives $8p^4 - 30p^2 = 0$, so $2p^2(4p^2 - 15) = 0$.

6. Find the possible a two values

If $p = 0$, then $a_2 = -7$. If $p^2 = \frac{15}{4}$, then $a_2 = 2 \cdot \frac{15}{4} - 7 = \frac{1}{2}$.

Final answer

$$(a) a_2 = 2p^2 - 7. \quad (b) a_2 = -7 \text{ or } \frac{1}{2}.$$

Question 9

Circles

9. A circle C has equation

$$(x - k)^2 + (y - 2k)^2 = k + 7$$

where k is a positive constant.

- (a) Write down, in terms of k ,

- (i) the coordinates of the centre of C ,
- (ii) the radius of C .

(2)

Given that the point $P(2, 3)$ lies on C

- (b) (i) show that $5k^2 - 17k + 6 = 0$
- (ii) hence find the possible values of k .

(3)

The tangent to the circle at P intersects the x -axis at point T .

Given that $k < 2$

- (c) calculate the exact area of triangle OPT .

(5)

Worked Solution - Question 9

1. Read the centre and radius

From $(x - k)^2 + (y - 2k)^2 = k + 7$, the centre is $(k, 2k)$ and the radius is $\sqrt{k + 7}$.

2. Substitute point P

Since $P(2, 3)$ lies on the circle, $(2 - k)^2 + (3 - 2k)^2 = k + 7$.

3. Show the quadratic

Expanding gives $k^2 - 4k + 4 + 4k^2 - 12k + 9 = k + 7$, hence $5k^2 - 17k + 6 = 0$.

4. Solve for k

$5k^2 - 17k + 6 = (5k - 2)(k - 3)$, so $k = \frac{2}{5}$ or $k = 3$.

5. Use the condition k less than two

For the tangent calculation, $k < 2$, so the centre is $\left(\frac{2}{5}, \frac{4}{5}\right)$.

6. Find the tangent gradient

The radius from the centre to P has gradient $\frac{3 - 4/5}{2 - 2/5} = \frac{11}{8}$, so the tangent gradient is $-\frac{8}{11}$.

7. Find the x intercept

The tangent is $y - 3 = -\frac{8}{11}(x - 2)$. Setting $y = 0$ gives $x = \frac{49}{8}$, so $T = \left(\frac{49}{8}, 0\right)$.

8. Calculate the area

Triangle *OPT* has base $\frac{49}{8}$ on the *x*-axis and height **3**, so its area is

$$\frac{1}{2} \cdot \frac{49}{8} \cdot 3 = \frac{147}{16}.$$

Final answer

$$(a)(i) (k, 2k). \quad (a)(ii) \sqrt{k+7}. \quad (b)(ii) k = 3 \text{ or } \frac{2}{5}. \quad (c) \frac{147}{16}.$$

Question 10

Modelling with Sequences & Series

10. In this question you must show detailed reasoning.

Owen wants to train for 12 weeks in preparation for running a marathon.

During the 12-week period he will run every Sunday and every Wednesday.

- On Sunday in week 1 he will run 15 km
- On Sunday in week 12 he will run 37 km

He considers two different 12-week training plans.

In training plan A , he will increase the distance he runs each Sunday by the same amount.

- (a) Calculate the distance he will run on Sunday in week 5 under training plan A . (3)

In training plan B , he will increase the distance he runs each Sunday by the same percentage.

- (b) Calculate the distance he will run on Sunday in week 5 under training plan B .
Give your answer in km to one decimal place. (3)

Owen will also run a fixed distance, x km, each Wednesday over the 12-week period.

Given that

- x is an integer
- the total distance that Owen will run on Sundays and Wednesdays over the 12 weeks will not exceed 360 km

- (c) (i) find the maximum value of x , if he uses training plan A ,
(ii) find the maximum value of x , if he uses training plan B . (5)

(Total 11 marks)

Worked Solution - Question 10

Topic group

1. Find the arithmetic difference

Plan A has 12 Sunday runs from 15 km to 37 km, so there are 11 equal increases.

The common difference is $\frac{37 - 15}{11} = 2$.

2. Find week five for plan A

Week 5 is the fifth term, so the distance is $15 + 4(2) = 23$ km.

3. Find the geometric ratio

For plan B, $15r^{11} = 37$, so $r = \left(\frac{37}{15}\right)^{1/11}$.

4. Find week five for plan B

The fifth Sunday distance is $15r^4 = 15\left(\frac{37}{15}\right)^{4/11} = 20.829\dots$, so it is 20.8 km.

5. Sum the Sunday runs for plan A

The arithmetic total is $\frac{12}{2}(15 + 37) = 312$ km.

6. Find the Wednesday limit for plan A

The total distance condition gives $312 + 12x \leq 360$, so $x \leq 4$. The maximum integer is 4.

7. Sum the Sunday runs for plan B

The geometric total is $15\frac{r^{12} - 1}{r - 1} = 294.185\dots$ km.

8. Find the Wednesday limit for plan B

Now $294.185\dots + 12x \leq 360$, so $x \leq 5.484\dots$. The maximum integer is 5.

Final answer

(a) 23 km. (b) 20.8 km. (c)(i) $x = 4$. (c)(ii) $x = 5$.